

OSTASKOV, Russian Federation

WMO: 263890

Lat: 57.133N

Lon: 33.117E

Elev: 218

StdP: 98.73

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.3	-22.0	-27.6	0.3	-25.0	-24.3	0.4	-21.6	6.6	-3.3	5.5	-4.9	0.8	320	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	28.3	19.6	26.3	18.8	24.5	17.9	20.7	26.4	19.7	24.6	18.7	23.3	1.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.9	23.0	17.8	13.1	22.2	16.9	12.4	21.1	60.5	26.6	57.2	24.6	54.1	23.3	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.6	5.2	4.7	-29.2	30.6	3.1	2.5	-31.5	32.4	-33.3	33.9	-35.0	35.3	-37.3	37.1		
(5)			-30.0	22.2	3.0	1.3	-32.2	23.2	-33.9	23.9	-35.6	24.7	-37.8	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-7.3	-7.1	-2.5	5.4	11.4	15.7	18.7	16.0	11.0	5.1	-0.7	-4.6	(6)
(7)		DBStd	10.25	6.96	6.99	4.89	4.59	4.40	3.56	3.04	3.32	3.42	4.03	5.06	6.22	(7)
(8)		HDD10.0	2552	536	478	387	150	36	1	0	1	28	160	322	452	(8)
(9)		HDD18.3	4887	794	712	646	387	217	97	32	88	222	412	572	711	(9)
(10)		CDD10.0	786	0	0	0	13	80	171	271	187	57	7	0	0	(10)
(11)		CDD18.3	79	0	0	0	0	3	16	44	16	1	0	0	0	(11)
(12)		CDH23.3	666	0	0	0	1	33	132	348	148	4	0	0	0	(12)
(13)		CDH26.7	153	0	0	0	0	5	26	83	39	0	0	0	0	(13)
(14)	Wind	WSAvg	2.0	2.1	2.1	2.2	2.0	2.1	1.8	1.5	1.6	1.8	2.1	2.2	2.2	(14)
(15)	Precipitation	PrecAvg	711	51	39	37	32	62	81	92	84	64	67	53	51	(15)
(16)		PrecMax	858	82	78	69	60	132	143	189	181	126	143	105	91	(16)
(17)		PrecMin	550	11	10	13	7	19	14	19	9	16	12	1	17	(17)
(18)		PrecStd	79	18	17	13	14	31	36	48	42	27	32	24	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	4.4	11.4	22.2	26.5	29.4	31.7	31.2	23.4	18.2	9.9	7.5	(19)
(20)			MCWB	3.3	2.9	5.8	12.2	17.5	19.2	21.6	20.6	17.0	13.1	8.9	6.8	(20)
(21)		2%	DB	2.1	2.9	7.9	19.3	23.7	26.8	28.8	26.8	20.5	13.8	8.0	5.3	(21)
(22)			MCWB	1.5	1.7	4.1	10.5	15.2	18.7	20.0	18.9	15.3	11.1	7.1	4.7	(22)
(23)		5%	DB	1.1	1.8	5.6	16.2	21.3	24.5	26.9	24.5	18.4	12.2	6.5	3.5	(23)
(24)			MCWB	0.5	0.8	2.9	9.7	14.1	17.6	19.4	18.0	14.1	10.5	5.7	2.9	(24)
(25)		10%	DB	0.5	1.0	3.7	13.7	19.1	22.4	25.1	22.2	16.6	10.8	5.2	2.1	(25)
(26)			MCWB	-0.1	0.3	1.8	8.4	12.8	16.4	18.5	16.9	13.3	9.4	4.5	1.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.5	7.0	13.7	18.4	20.9	22.5	21.8	18.0	13.6	9.1	6.9	(27)
(28)			MCDWB	3.7	4.2	10.3	21.0	25.6	26.7	28.6	29.7	22.0	16.6	9.5	7.4	(28)
(29)		2%	WB	1.7	1.9	4.6	11.5	16.6	19.6	21.1	20.1	16.2	11.8	7.0	4.7	(29)
(30)			MCDWB	1.9	2.5	6.7	17.1	21.6	24.5	26.9	25.0	19.4	13.2	7.6	5.2	(30)
(31)		5%	WB	0.7	1.1	3.2	10.1	15.1	18.6	20.1	18.7	14.8	10.7	5.7	3.0	(31)
(32)			MCDWB	0.9	1.6	5.1	15.4	19.5	23.5	25.5	22.7	17.5	12.2	6.3	3.5	(32)
(33)		10%	WB	0.0	0.4	2.1	8.9	13.8	17.3	19.2	17.6	13.7	9.5	4.5	1.6	(33)
(34)			MCDWB	0.4	1.0	3.5	13.3	17.7	21.3	23.9	21.4	16.2	10.8	5.1	2.0	(34)
(35)	Mean Daily Temperature Range	MDBR		4.9	6.5	7.7	9.4	9.5	8.6	9.3	8.6	7.7	5.2	3.8	4.0	(35)
(36)		5% DB	MCDBR	3.5	4.3	10.0	14.2	13.3	12.0	12.7	12.9	11.2	7.5	4.7	3.5	(36)
(37)			MCWBR	3.5	3.6	6.6	7.4	6.2	5.1	5.3	5.9	6.4	5.1	4.2	3.4	(37)
(38)			5% WB	MCDBR	3.5	3.5	7.9	12.4	11.8	10.5	11.4	10.6	9.4	6.6	4.2	3.5
(39)		MCWBR		3.6	3.2	5.6	7.1	6.0	5.1	5.4	5.5	6.1	4.8	4.0	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.280	0.314	0.335	0.384	0.377	0.366	0.393	0.398	0.359	0.340	0.314	0.273	(40)
(41)		taud		2.135	2.114	2.174	2.250	2.378	2.431	2.366	2.342	2.425	2.398	2.241	2.133	(41)
(42)		Ebn at Noon		574	711	808	826	860	875	839	805	787	696	534	471	(42)
(43)		Edh at Noon		55	87	107	117	110	105	110	106	84	67	51	42	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.44	1.23	2.72	3.96	5.13	5.29	5.17	4.12	2.62	1.15	0.40	0.25	(44)
(45)		RadStd		0.10	0.20	0.29	0.43	0.47	0.53	0.62	0.42	0.32	0.24	0.07	0.07	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page